UNIT 2 · TOPIC 2(B)

Couplings and Overload-Protection Devices

Syllabus topic: Design of bearings, couplings, springs and fasteners

Couplings

Coupling	Accommodates
Rigid — muff/sleeve, flange	Perfectly aligned shafts only
Flexible — pin-bush, jaw (spider), tyre	Small misalignment + shock damping
Oldham	Parallel offset
Universal (Hooke's) joint	Angular misalignment

Hooke's joint speed fluctuation

- A single Hooke's joint at angle α makes the driven-shaft speed fluctuate once per revolution, between **cos α** and **1/cos α** times the input speed.
- Cancel the fluctuation with **two joints, yokes in phase and equal joint angles** (as in PTO drivelines), or use CV joints (wide-angle PTO).

Overload safety devices (Kepner §4.19–4.21)

Device	How it works / where used
Shear pin/bolt	Breaks at a set overload (baler flywheel, rotavator, forage harvester); replace it after each overload.
Slip (friction) clutch	Spring-loaded friction discs slip at a set torque and reset automatically; protects against frequent momentary overloads (PTO drives of rotavators, balers).
Jump (ratchet/cam) clutch	Cams jump with a loud clatter at overload and re-engage when the load drops — an audible warning.
Overrunning (freewheel) clutch	Transmits torque one way only; lets implement inertia (rotavator, mower flywheel) overrun the tractor PTO during deceleration — a safety must on PTO machines with high-inertia rotors.

Practice MCQs

Q32. Speed fluctuation of a single Hooke's joint is eliminated in a PTO shaft by:

(a) Higher speed
(b) Two joints with in-phase yokes and equal joint angles
(c) Heavier shaft
(d) Lubrication

Q33. Oldham coupling accommodates:

(a) Angular misalignment
(b) Parallel (lateral) shaft offset
(c) Axial float only
(d) Torsional vibration

Q34. The overload device that gives an audible warning and re-engages automatically is the:

(a) Shear pin
(b) Jump (ratchet) clutch
(c) Rigid coupling
(d) Key

Q35. An overrunning (freewheel) clutch in a rotavator/PTO driveline:

(a) Limits torque
(b) Allows the high-inertia rotor to overrun the PTO while slowing
(c) Reverses drive
(d) Increases speed

Answers

Q32 — (b) Two joints with in-phase yokes and equal joint angles — *PTO drivelines are double-Cardan arrangements; equal angles + in-phase yokes cancel the cos α fluctuation.*

Q33 — (b) Parallel (lateral) shaft offset

Q34 — (b) Jump (ratchet) clutch

Q35 — (b) Allows the high-inertia rotor to overrun the PTO while slowing

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.